Clarissa Morgan

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Education

California State University, Fresno — Fresno, CA
B.S. Computer Engineering | GPA: 3.7 | President's List (2 semesters) — *Aug. 2024 – Expected Dec. 2027*

Lemoore College & Lemoore Middle College High School — Lemoore, CA
Earned concurrently: A.A. Mathematics & Science, A.A. Arts & Humanities, H.S. Diploma — *Aug. 2020 – May 2024*

Experience

Computer Engineering Intern — May 2026 – Aug. 2026
Intel Corporation — *Hillsboro, OR*

- Built an AI analytics system (Python, SQL, Polars) that analyzes fab data through Intel's internal LLM gateway, generating risk analysis, predictions, and insights; cut an estimated 75% of manual analysis work
- Made it work with any database and split large datasets (1,000+ columns, 10,000+ rows) into chunks the model processes reliably, keeping all data inside Intel's network (OAuth2, SSL verification)
- Built a tool-inventory dashboard (Power BI, TOAD Data Point, Docker) saving its team ∼7 hrs/week, and a parallelized utility that resolves FQDNs and pings fab hosts in batches, saving ∼2 hrs/week
- Prototyped RAG and MCP integrations with internal AI tools; documented the RAG setup for other employees

Automotive Ethernet Research Assistant — Jan. 2026 – Present
California State University, Fresno — *Fresno, CA*

- Characterize voltage profiles of automotive Ethernet networks and develop cybersecurity measures for them

AI & Robotics Research Assistant — May 2025 – July 2025
California State University, Fresno — *Fresno, CA*

- Upgraded a 6-DoF strawberry-harvesting arm's vision (Python, NVIDIA Jetson AGX Orin) from 2D color detection to 3D detection that measures each strawberry's distance so inverse kinematics can guide the arm
- Discovered the arm's camera stream stayed accessible while powered off; documented the vulnerability

Projects

Hybrid Counting Sort / Frequency Red-Black Tree | *C++* — Feb. 2026 – May 2026

- Co-designed a sort that switches between counting sort and a frequency-counting red-black tree by value range: $O(n)$ time when range $\leq 100n$, otherwise $O(n \log k)$ for k distinct values
- Benchmarked 8 input types up to 1.3M elements: 64% faster and 93% less memory than red-black tree sort

Network Attack Lab: DoS & Packet Replay | *hping3, nginx, VirtualBox, Linux* — Jan. 2026 – Present

- Ran and analyzed packet-replay and DoS port-flooding attacks on a Minecraft server in an isolated VM lab

AES-128 Core with Hardware Trojan | *Verilog, FPGA* — Fall 2026

- Co-designed a synthesizable AES-128 core in Verilog; verified encryption output against NIST test vectors
- Inserted a hardware trojan that corrupts output only on a hidden trigger; verified in simulation and on FPGA

Nios V (RISC-V) Embedded System | *C, Quartus Prime Pro, Platform Designer* — Course Project

- Built a Nios V system with peripherals; programmed board I/O in C using timers and interrupts

Leadership & Teaching

Chair (President) — Jan. 2025 – Present
IEEE Fresno State Student Branch — *Fresno, CA*

- Grew events from ∼5 to 30+ per semester, including PCB, soldering, and Python workshops for 50+ students
- Re-established ties with our IEEE section and area, then secured $6,000 to send 11 members to IEEE Rising Stars
- Brought Fresno State into the IEEE Elevate pilot, funding senior capstone work at an under-resourced campus, and revived the Micromouse robot competition in partnership with the Robotics Club

Founder & President | *Lemoore Middle College High School Peer Tutoring Program* — Apr. 2021 – May 2024

- Founded a peer tutoring program and led a team of 4 tutors in delivering 500+ hours of academic support
- Raised tutored students' grades by 1–2 letter grades on average, helping at-risk students recover from failing

Technical Skills

Languages: C, C++, Python, SQL (PostgreSQL), Verilog, ARM Assembly, MATLAB, PowerShell
Hardware: Nios V (RISC-V), FPGAs (DE1-SoC, DE25), Jetson, Raspberry Pi, Vivado, ModelSim, oscilloscopes
Security: Threat modeling (STRIDE, DREAD), OWASP IoT, Linux hardening
AI & Data: LLM integration, RAG, MCP, Polars, pandas, Power BI (DAX, Power Query), TOAD Data Point, cnvrg.io
Systems & Networking: Linux, Git, Docker, VirtualBox, Tailscale, SSH, nginx, nmap, hping3, OAuth2